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Reflection Paper #3

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Twigs Naturals

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Internship Reflection Paper

Third 50 Hours

The third 50 hours in my cybersecurity internship with Twigs Naturals has marked a transition from mainly supporting governance, risk, and compliance (GRC) tasks to taking on more independent analysis and security planning responsibilities. While some of my earlier work focused on monitoring training completion, managing access controls, and helping with documentation, I've recently contributed more directly by identifying security gaps and creating more structured solutions. As a result, I've been able to better understand how my GRC efforts have gradually progressed into more actionable security strategies that will help support my host company's cybersecurity posture.

One the most important projects I've worked on during this phase included developing an Email Security Enhancement Plan that will be used to implement SPF, DKIM, and DMARC for the organization's domain. This project involved investigating the company's pre-existing email infrastructure, determining the presence or absence of specific authentication mechanisms, and

assessing the possible risks of absent mechanisms, including spoofing and phishing. I also analyzed how various systems, such as Google Workspace, WordPress, and HubSpot, work together in the organization's domain. Based on these findings, I helped design a phased implementation plan, which includes monitoring procedures before enforcing changes, to avoid disrupting operations. This helped me understand the importance of securing communication channels, and how important policy-driven configurations are for defending company assets.

Another project focused on documenting and evaluating access controls for the company software based in AppSheet. This was an opportunity to collaborate with another intern to gather information about user roles, permissions, and system behavior across various apps and each of their module instances. I helped formally document and explain the access control model that my supervisor wanted for all company software in AppSheet for the long-term, which would help the company save money by intentionally (but carefully) exposing the software to public access. This will require Google authentication, and it will allow scalability beyond the default user limitations. By controlling access within the applications instead of at the entry point, the software can remain secure as long as the applications utilize the documented access control logic. Ultimately, I also helped with testing access by using both authorized and unauthorized accounts to validate permission enforcement. This helped me realize the importance of ensuring that implemented controls align with documented security policies, and determining if there are any gaps between design and real-world usage.

I've also contributed to the creation of an archival and data integrity framework that was part of a cross-departmental customer service improvement initiative. This involved helping define structured approaches for organizing and retaining data, such as proposing a naming convention for archived files, access permissions for various employee/intern roles, and retention

timelines for various types of organizational information. My role in this project has caused me to gain an appreciation for how data governance ensures that information is correctly managed for its intended lifecycle. This experience taught me how keeping organized, accurate, and accessible records is important for decreasing risk and supporting compliance efforts.

As I reflect on my internship as a Cybersecurity Technician Intern, I have developed a deeper understanding of how GRC functions affect the overall cybersecurity posture of an organization. My experience has taught me that cybersecurity efforts are not exclusively dependent on technical tools; instead, cybersecurity is also about maintaining well-defined policies, organized processes, and consistent user accountability. While this work has been both interesting and challenging along the way, it's still meaningful to me, and it's rewarding to see my efforts contribute to the improvement of real-world security practices. Additionally, I recognize that my organizational skills, attention to detail, and communication skills align with roles in this area of cybersecurity. Overall, my internship has enhanced my interest in pursuing a career that incorporates both technical aspects and governance aspects of cybersecurity, and I'm looking forward to building upon these skills after my time with Old Dominion University and Twigs Naturals as well.